

Objects

Introduction to Programming

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Introduction

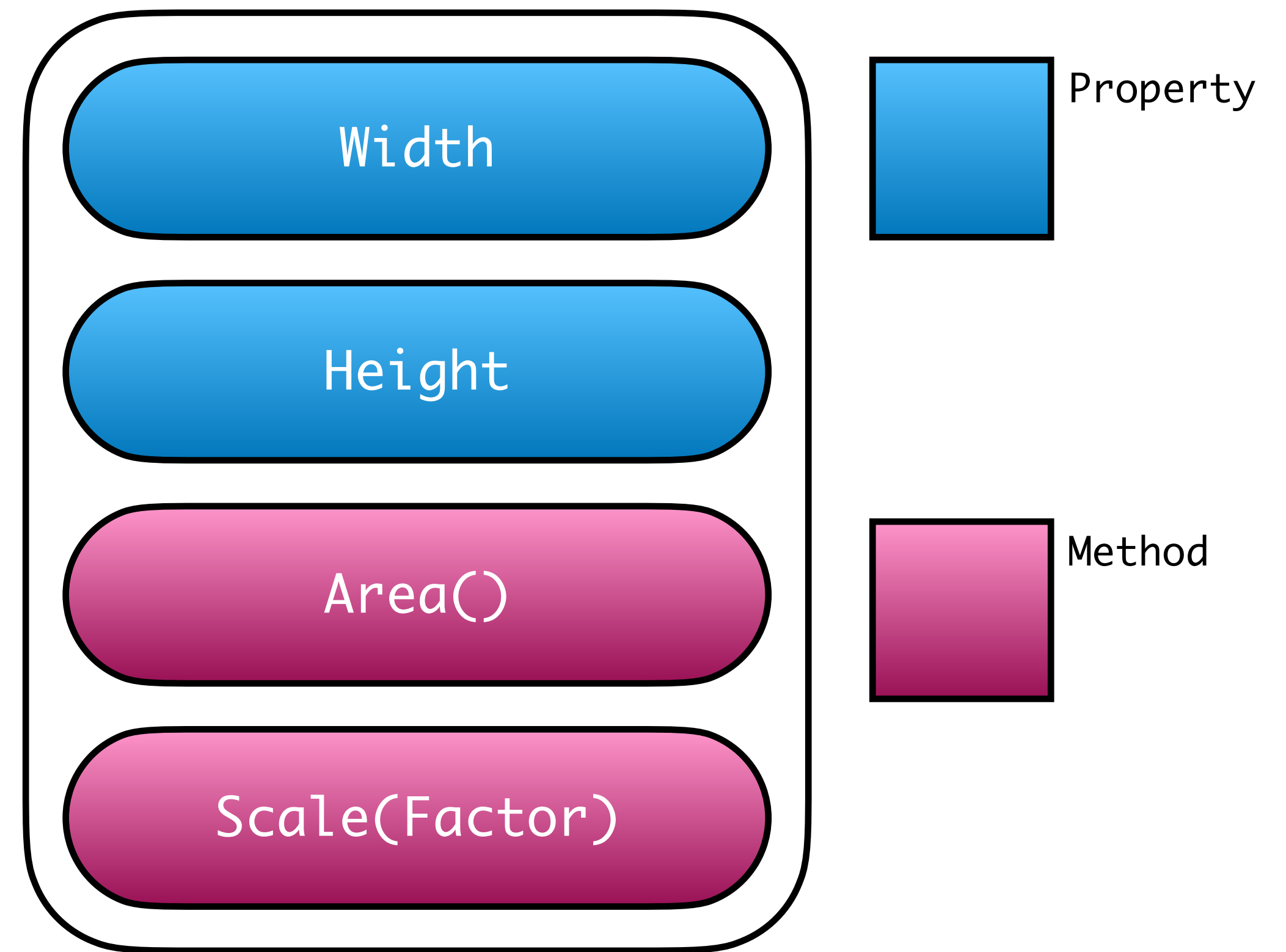
Objects

- **Object:** a group of related variables and functions
- Variables in an object are called attributes/fields/properties
- Functions in an object are called methods
- An object's methods should use/change its properties

Introduction

Objects

- A rectangle has a width & height
 - These are its properties
 - An object's props and their values represent the object's state
- A rectangle's area is computed using its props (so area must be a method)
- The scale method updates the object's props



Classes

Objects

- **Class:** a template for creating objects; a data type
- The class dictates what properties & methods each object of its type has
- The class does NOT set the values of its properties (each object is different)

Classes

Objects

- `class` Rectangle:
- `def __init__(self, width, height):`
- `self.width = width`
- `self.height = height`

Object Creation

Objects

- `rect1 = Rectangle(6, 4)`

Getting Properties

Objects

- `print(rect1.width)`

Setting Properties

Objects

- `rect1.width = 8`

Declaring Methods

Objects

- `class Rectangle:`
- `...`
- `def area(self):`
- `return self.width * self.height`

Calling Methods

Objects

- `print(rect1.area())`

Object Oriented Programming

Objects

- An approach to program design & development focused on things
- The program's functionality is grouped into objects alongside the data said functions operate upon
- Problem description nouns might indicate classes